5 Mass of water sends back per sec, $m = (vol)\rho = (\pi r^2 v)\rho$

power of motor = 2 times gain in K.E. of water per sec

$$= 2\left(\frac{1}{2}mv^2\right)$$

$$= 2\left(\frac{1}{2}\left(\left(\pi r^2 v\right)\rho\right)v^2\right) = 31.4 \text{ kW}$$

Answer: D

6 The skateboarder will experience the maximum force by the half pipe at the most bottom